

Define Deployment Phase SDLC

The deployment phase is the stage where fully developed, tested, and validated software is released to the production environment for end-user access. It involves final configuration, infrastructure setup, and the official launch of the application.

Deployment Models

- **Manual Deployment:** Human-led installation of software, file transfers, and server updates. It allows direct control but is prone to human error and inconsistency.
- **Continuous Deployment (CD):** An automated process where code that passes all automated tests is automatically pushed to production. This ensures consistent, repeatable, and fast releases (Continuous Delivery Foundation, 2023).

Tools and Platforms

1. **Jenkins:** An open-source automation server used to build, test, and orchestrate CI/CD pipelines (Docker, 2025).
2. **AWS (Amazon Web Services):** Provides cloud infrastructure (e.g., EC2) and release management tools (e.g., CodePipeline) to host and deploy applications.
3. **Docker:** A containerization platform that packages applications and dependencies together to ensure they run identically across all environments (Docker, 2025).

The Role of DevOps

DevOps practices bridge the gap between development and operations through automation and feedback loops. By utilizing Infrastructure as Code (IaC) and CI/CD pipelines, teams minimize configuration errors, reduce deployment risk, and improve overall system reliability (Microsoft, 2026).

Real-World Incident

In 2012, Knight Capital Group suffered a \$460 million loss in 45 minutes due to a deployment error. Technicians failed to update one of eight production servers, causing dormant legacy code to be triggered by the new software, which resulted in massive erroneous stock trades. This incident serves as a critical case study on the necessity of rigorous deployment checklists and automated verification (PRMIA, 2026).

References

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